

WHITEPAPER

Decentralized Personal Retirement Fund on Blockchain

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Cristian F. Taborda | Eternal Foundation Team

contact@eternal.fund

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Executive Summary

Eternal Fund Protocol is a decentralized personal retirement fund system built on blockchain (Arbitrum / Polygon), enabling anyone in the world to create and manage their own retirement fund autonomously, transparently, and without traditional financial intermediaries.

The protocol combines smart contracts written in Vyper with integrations to established DeFi protocols (Aave v3) and tokenized real-world assets (Ondo USYD / US T-Bills) to offer secure yields on USDC deposits. All business logic is encoded in auditable contracts, and funds always remain under the user's control.

Parameter	Value
Core contract	Vyper ^0.4.3 (PersonalFund, Factory, Treasury, Registry)
Deposit currency	USDC (6 decimals)
Management fee	5% per deposit (adjustable, maximum 10%)
Integrated yield	Aave v3 (DeFi) + Ondo USYD (RWA / T-Bills)
Fund timelock	15 to 50 years (user-configurable)
Utility token	ETRF — 1,000,000,000 total supply
Core contract license	BUSL-1.1 → GPL-2.0+ in 2030

Eternal's goal is to democratize access to retirement savings, offering a transparent and non-custodial alternative to traditional pension systems, accessible globally with just a wallet and stablecoins.

1. Problem and Solution

1.1 The Global Retirement Problem

Traditional pension systems face a structural crisis worldwide: pension funds with growing deficits, inflation eroding purchasing power, and increasing labor informality leaving millions of workers without pension coverage. In Latin America, less than 40% of workers actively contribute to formal pension systems.

Additionally, conventional savings instruments (fixed deposits, mutual funds) present significant barriers to entry: high minimums, opaque fees, dependence on intermediaries, and geographically restricted access.

1.2 The DeFi Opportunity

Blockchain technology and decentralized finance (DeFi) protocols enable, for the first time, the construction of long-term financial instruments that are:

- Non-custodial: users maintain control of their funds at all times.
- Transparent: all logic is encoded in public, verifiable contracts.
- Borderless: accessible from anywhere in the world with an internet connection.
- Programmable: saving, investment, and withdrawal rules execute automatically.
- Composable: integrate established DeFi protocols to generate yield.

1.3 The Solution: Eternal Fund Protocol

Eternal Fund Protocol resolves these limitations through a set of smart contracts that allow each user to create their own personal retirement fund on blockchain, with customized parameters (retirement age, monthly deposit, risk tolerance, preferred investment protocol) without ceding control of their assets to any third party.

2. Protocol Architecture

The protocol consists of a set of interdependent smart contracts, organized in functional layers. All core contracts are written in Vyper ^0.4.3 for its superior security and auditability compared to Solidity.

2.1 Core Contracts (BUSL-1.1)

PersonalFund.vy — User Personal Fund

The central contract of the protocol. Each user has their own deployed instance (minimal proxy EIP-1167). It manages the complete fund lifecycle: initial deposits, monthly deposits, DeFi protocol investments, retirement withdrawal, and emergency withdrawal.

PersonalFundFactory.vy — Fund Factory

Deploys new PersonalFund instances using the minimal proxy pattern (EIP-1167 / clones), significantly reducing gas costs. Validates all input parameters before deployment and registers created funds.

Treasury.vy — Protocol Treasury

Manages the collection and custody of protocol fees, and acts as arbiter for early withdrawal requests. Default fee: 5% (DEFAULT_FEE = 500 basis points). Maximum fee: 10% (MAX_FEE = 1000 basis points).

ProtocolRegistry.vy — Protocol Registry

Curated catalog of DeFi protocols and tokenized assets available for investment. Maintains metadata for each protocol: name, category, estimated APY, risk level, accumulated TVL, and verification status.

UserPreferences.vy — User Preferences

Manages individual user configuration: selected DeFi protocol, auto-compound activation, risk tolerance, and deposit routing strategy.

2.2 DeFi Adapters (BUSL-1.1)

AaveV3Adapter.vy — Aave v3 Integration

Bridge between the Eternal protocol and the Aave v3 lending market on Arbitrum One. Implements the safe allowance pattern (reset to 0 before each approval). Contract on Arbitrum One: Pool 0x794a61358D6845594F94dc1DB02A252b5b4814aD.

OndoUSDYAdapter.vy — Ondo Finance USDY Integration

Adapter for investing in USDY, a tokenized promissory note backed by short-term US Treasury bills. Historical APY: 4.5%–5.5%. Implements the secondary market route via Uniswap v3 (USDC → USDY). Estimated slippage: 0.01%–0.05%.

3. Personal Fund Lifecycle

3.1 Fund Creation

Parameter	Description
principal	Initial seed capital in USDC (0 – 100,000 USDC)
monthlyDeposit	Fixed amount to deposit every 30 days
currentAge	User's current age (validated by factory ranges)
retirementAge	Target retirement age (> currentAge)
desiredMonthly	Desired monthly amount during retirement
yearsPayments	Years of payments in the retirement phase (1–50)
timelockYears	Years until retirement / fund lockup period (15–50)
selectedProtocol	Initial DeFi adapter address (must be approved)

3.2 Accumulation Phase

During the lockup period (timelockYears), the user makes periodic monthly deposits and accumulates yield from DeFi protocols. The protocol records missed months (missedMonths) informatively without applying penalties. Extra deposits have a 24-hour window (EXTRA_GRACE_PERIOD) to be rescinded with a 1% penalty.

3.3 Early Withdrawal

If the user needs access to their funds before the timelock expires, they must request approval from the protocol Treasury via requestEarlyRetirement(reason). The admin can approve or reject the request.

3.4 Emergency Withdrawal

In extreme situations, the admin can announce an emergency withdrawal via announceEmergency(). There is a mandatory waiting period of 48 hours (EMERGENCY_DELAY = 172,800 seconds) before the user can execute emergencyWithdraw().

3.5 Retirement Phase

Upon completion of the timelock period, the owner activates the retirement phase with retirement(). The system automatically configures monthly auto-withdrawals for the desired amount (desiredMonthly). The minimum interval between withdrawals is 7 days (604,800 seconds).

4. ETRF Token — Eternal Fund Token

UTILITY DECLARATION: ETRF is exclusively a utility token. It does NOT represent equity, debt, dividends, or any expectation of profit derived from the work of third parties. It is not designed as an investment instrument or security under any jurisdiction.

4.1 Token Use Cases

- Premium Mode: users holding a minimum of ETRF unlock advanced application features (reports, investment strategies, alerts). The token is only verified, not consumed.
- Fee Discount: users can pay management fees in ETRF instead of USDC, receiving a discount. ETRF tokens used are burned, reducing the circulating supply.
- Prepaid Credits: users acquire ETRF in advance and the protocol gradually burns them as usage fees are generated.

4.2 Tokenomics

Total supply: 1,000,000,000 ETRF (fixed, non-mintable post-deploy).

Allocation	Percentage	Tokens (ETRF)	Details
Public Sale	30%	300,000,000	Distributed by rounds via SaleETRF → VestingETRF
Team	20%	200,000,000	1-year cliff + 25% unlock + 75% linear over 3 years
Ecosystem	20%	200,000,000	Incentives, partnerships, integrations
Treasury	15%	150,000,000	Receives USDC from sales and unsold tokens
Community	10%	100,000,000	Community programs, rewards
Reserve	5%	50,000,000	Strategic reserve

4.3 Vesting Mechanism

Public sale tokens are distributed linearly after a round-configurable cliff period. Team vesting: 1-year cliff → 25% instant unlock → 75% linear over the following 3 years.

5. Security and Risk Management

5.1 Why Vyper

All core contracts are written in Vyper, a smart contract language designed with an emphasis on security and auditability. It prohibits multiple inheritance, operator overloading, recursion, and inline modifiers, significantly reducing attack surface.

5.2 Access Controls

Control	Mechanism
Fund owner	Only the PersonalFund owner can make deposits and withdrawals
Protocol admin	Separate role for administrative operations (Treasury, Factory, Registry)

Control	Mechanism
Protocol whitelist	Only active and verified protocols in ProtocolRegistry are usable
Fund registry	Only funds registered in Factory can report fees to Treasury

5.3 Risks to Consider

The following risks are inherent to DeFi protocols and blockchain usage. Users must understand them before participating.

- Smart contract risk: no code is 100% bug-free. Funds may be lost in the event of an exploit.
- DeFi protocol risk: Aave v3 and Ondo USDY are external. An exploit there could affect deposited funds.
- Stablecoin depeg risk: USDC may lose its USD parity under extreme circumstances.
- Regulatory risk: the legal framework for digital assets is uncertain and varies by jurisdiction.
- Liquidity risk: in market stress, liquidity in Uniswap v3 USDY/USDC pools could decrease.
- Admin risk: dependence on the Ethereum Foundation for key administrative operations.

6. Legal and Regulatory Framework

6.1 Contract Licensing

Contract	License	Term
PersonalFund.vy	BUSL-1.1	Until 2030-01-07 → GPL-2.0+
PersonalFundFactory.vy	BUSL-1.1	Until 2030-01-01 → GPL-2.0+
Treasury.vy	BUSL-1.1	Until 2030-01-01 → GPL-2.0+
AaveV3Adapter.vy	BUSL-1.1	Until 2030-01-01 → GPL-2.0+
OndoUSDYAdapter.vy	BUSL-1.1	Until 2030-01-01 → GPL-2.0+
ProtocolRegistry.vy	Apache-2.0	Perpetual
UserPreferences.vy	Apache-2.0	Perpetual
DateTime.vy	Apache-2.0	Perpetual
ETRF.vy	MIT	Perpetual
SaleETRF.vy	MIT	Perpetual
VestingETRF.vy	MIT	Perpetual

6.2 Token Classification

The ETRF token has been designed as a pure utility token. Its value derives solely from its use within the Ethereum Fund ecosystem and not from any expectation of profit. Each user is responsible for determining the legal and tax classification of the ETRF token in their jurisdiction.

7. Roadmap

Phase	Milestones
Q2 2026 — Testnet	Full deployment on Arbitrum Sepolia. Security audit. Bug bounty program.
Q3 2026 — Mainnet v1	Launch on Arbitrum One. ETRF public sale (Seed Round). First user funds.
Q4 2026 — Expansion	Integration of new DeFi protocols. Deployment on Polygon. More RWA adapters.
2027 — Decentralization	On-chain governance for Protocol Registry. Reduction of admin privileges.
2030 — Open Source	Migration from BUSL-1.1 to GPL-2.0+ licenses. Fully open-source protocol.

8. Team and Contact

Lead Architect: Cristian F. Taborda — Eternal Foundation

Email: contact@eternal.fund | **Legal:** legal@eternal.foundation

Website: eternal.fund | **Docs:** docs.eternal.fund (coming soon)

9. Legal Disclaimer

This whitepaper has been prepared solely for informational purposes about the Eternal Fund Protocol and the ETRF token. It does not constitute, nor should it be interpreted as, an offer or solicitation to purchase or subscribe to securities, financial assets, or other instruments in any jurisdiction.

Investment in cryptocurrencies and DeFi protocols entails substantial risks, including the possibility of total loss of invested capital. The Eternal Foundation accepts no liability for direct, indirect, incidental, or consequential losses.

This document may contain forward-looking statements based on the team's current expectations. The roadmap is subject to change without notice.